Permafrost Is Thawing From the Inside Out in Canada's North

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Deep in Canada's Northwest Territories, the ground is changing. **Permafrost**, ground that stays frozen for at least two years in a row, is thawing. In some places, it is melting from the inside out and setting off a new kind of landslide. The Mackenzie River winds through this frozen landscape. It is North America's largest northward-flowing river and drains about 20 percent of Canada's land. For thousands of years, **permafrost** held the region steady, but warming is crumbling that stability. In 2017 and 2018, two landslides struck the central Mackenzie Valley, moving 4.5 and 18 million **cubic meters** of frozen ground. The landslides carried chunks of earth that were still frozen, which had slid hundreds of meters. In 2019, **geologist** Joseph Young saw slopes collapsing with landslides everywhere. His team tracked ground temperatures from 2020 to 2022. They concluded that wildfires from decades earlier and rising summer temperatures had formed cracks in the ground. Water **seeped** in through the cracks and warmed the **permafrost** from below, weakening its base. The researchers named the new phenomenon **permafrost** detachment slides**.**

Word Treasure

permafrost -- Ground that stays frozen for at least two years in a row.

geologist -- A scientist who studies rocks and the ground.

cubic meters -- A way to measure how much space something takes up.

seeped -- Moved slowly through small cracks or gaps.

permafrost detachment slides -- A new kind of landslide where frozen ground slides away from a slope.

Background you need

Permafrost is ground in cold northern places that stays frozen for years, holding the land steady like frozen glue.

In Canada's far north, the Mackenzie River winds through this frozen land, and the ground there has held firm for a very long time.

When the ground warms up, cracks can form, water can seep in, and the frozen ground can weaken and slide away.

Break it down

WHO: Geologist Joseph Young and his team of researchers.
WHAT: The frozen ground is melting in a new way and setting off landslides.
WHERE: In Canada's Northwest Territories, around the central Mackenzie Valley.
WHY: Wildfires from decades earlier and rising summer temperatures made cracks, and water seeping through the cracks warmed the permafrost from below.

Why it matters

Learning how frozen ground can melt and slide helps us understand why the places where people live and travel in the far north are changing as the Earth warms.

Test what you remember

1. What is permafrost?

- (A) A river that flows north
- (B) Ground that stays frozen for at least two years in a row
- (C) A kind of wildfire
- (D) A team of scientists

2. Where did the two big landslides happen?

- (A) Inside a city park
- (B) In the central Mackenzie Valley
- (C) At the top of a volcano
- (D) On a beach far from Canada

3. What did Joseph Young and his team say made cracks in the ground?

- (A) Wildfires from decades earlier and rising summer temperatures
- (B) Earthquakes under the river
- (C) Animals digging in the ground
- (D) Heavy rain in one week

4. How did water make the permafrost weaker?

- (A) It seeped through cracks and warmed the permafrost from below
- (B) It washed the whole river away
- (C) It froze the ground even harder
- (D) It turned the ground into solid ice

5. What did the researchers name the new kind of landslide?

- (A) Summer ground slides
- (B) Permafrost detachment slides
- (C) Frozen river slides
- (D) Mackenzie River slides

6. How much of Canada's land does the Mackenzie River drain?

- (A) About 50 percent
- (B) About 20 percent
- (C) About 5 percent
- (D) About 90 percent

Answer key (try the quiz first!): 1: B 2: B 3: A 4: A 5: B 6: B

Questions to think about

MY THOUGHTS (at least 20 words):